

Bhavik Barot

6062 Avenue McDonald, Montreal, QC H3X 2X1

☎ 438-979-4921 — ✉ bhaviksbarot@gmail.com — [in LinkedIn](#) — [GitHub](#)

Summary

Mechanical / Aerospace Engineer (M.Eng., Concordia University) with graduate-level training in stress, fatigue, and thermal analysis using ANSYS Workbench/APDL. Experience in refinery equipment design under ASME guidelines and aerospace manufacturing at Safran Aerospace and Pratt & Whitney Canada. Skilled in applied combustion and CFD for gas turbine engines. Familiar with piping stress analysis, design documentation (GA drawings, isometrics, BOMs), and exposure to ASME/CSA codes. Eligible for PEO-EIT and motivated to build expertise in nuclear piping analysis.

Skills

- **Programming & Data Analysis:** Python, MATLAB, Cantera (Combustion Modeling), Power BI, VBA, MySQL
- **Simulation & CAD Tools:** AutoCAD, Solidworks, CATIA V5 (3-D Modeling), Siemens NX, ANSYS (Fluent, Structural, Classic), StarCCM+, Nastran, Patran, ANSYS APDL, Simulink
- **Collaboration & Productivity Tools:** Microsoft Office, Power BI, SharePoint, SAP, Enovia, ESYNC, Jira
- **Codes & Standards:** Familiar with ASME BPVC Sec VIII, exposure to ASME Sec III, B31.1, CSA B51/N285.0.
- **Languages:** English, French (TEFAQ-B2 Level)

Experience

Safran Aerospace - Kirkland, Quebec

March - 2025

Method Designer

- Leading a project to automate manufacturing work instruction (MWI) development by detecting iterative design changes using Python-based scripts, enhancing traceability and reducing manual rework.
- Integrated configuration management practices by extracting design data from databases and comparing with previous iterations to identify key changes impacting manufacturability.
- Collaborated with cross-functional teams (design, engineering, and production) to ensure accurate interpretation of Condition of Supply (CSD), BOMs, and 3D/2D models using CATIA V5 and Composer 3DVia.

Pratt & Whitney Canada - Longueuil, Quebec

May 2024 – Sept 2024

Operations Program Analyst (Full-Time Internship)

- Supported engine repair and overhaul activities by ensuring materials met high-performance standards. Collaborated with cross-functional teams to address material-related challenges and ensured alignment with design specifications.
- Developed a Project Management Dashboard POWER BI and Power Automate, automating processes and enabling real-time tracking to support operational excellence through Continuous Improvement.
- Optimized process flow by aligning Management and Engineering BOMs, improving efficiency by 15% and enabling accurate engine cost evaluations while providing reports to stakeholders. Supported sourcing and asset management of engineering components while enhancing overall customer satisfaction.

Vitthalesh Engineering -Gujarat, India

Aug 2021 – Aug 2022

Graduate Engineer – Mechanical Design & Simulation

- Modeled refinery pressure equipment and structural supports in SolidWorks, including vessels, pipe racks, and skid-mounted assemblies supplied to Larsen & Toubro (L&T) for oil & gas projects. Prepared General Arrangement (GA) drawings, fabrication layouts, and bills of materials (BOMs) for manufacturing release.
- Conducted stress and fatigue analysis in ANSYS Workbench/APDL on support brackets, nozzle connections, and welded joints. Defined boundary conditions, internal pressure loads, and thermal cases; prepared calculation reports with assumptions, load cases, and validation checks.
- Collaborated with QA and manufacturing teams to address non-conformance reports (NCRs) and design deviations during fabrication. Supported root-cause analysis and updated documentation to maintain compliance with ASME Boiler & Pressure Vessel Code (Sec. VIII) requirements.

Larsen & Toubro Defence IC -Gujarat, India

Jan 2021 – Aug 2021

Mechanical Design Intern

- Designed 3D models and 2D drawings in CATIA V5 as per GD&T and supportability standards, collaborated with production to ensure manufacturability, compliance, and efficient logistics through transportation considerations.
- Applied Model-Based Systems Engineering (MBSE) to define and communicate system architectures for accessory component integration, improving coordination across design, analysis, and manufacturing in the product lifecycle.

Education

Concordia University

Sept 2022 – Sept 2024

M.Eng Aerospace Engineering

Major : Combustion, Thermofluid Science

Gujarat Technological University

July 2017 – June 2021

B.E. Mechanical Engineering

Major : Mechanical Engineering

Projects

Structural Integrity Assessment of Pressure-Induced Stresses in Piping Elbows Using ANSYS APDL

- **Objective:** To model a 90° pipe elbow under internal pressure and evaluate hoop and axial stresses; verify that stresses are below allowable limits per ASME B31.1. Also perform mesh refinement to ensure accuracy.
- **Solution:** Create geometry for a 90° elbow in ANSYS APDL. Assign material (e.g. carbon steel) with relevant mechanical properties. Apply internal pressure load. Constrain ends (e.g. fixed or pinned) to prevent rigid body motion. Use solid or shell elements. Refine mesh near bends. Extract hoop (circumferential) and axial (longitudinal) stress values. Compare results with ASME code allowable stresses.
- **Result:** The 90° pipe elbow was analyzed under internal pressure using ANSYS APDL. Stresses were extracted at both the intrados (inner curvature) and extrados (outer curvature) of the bend. Results showed higher tensile hoop stresses at the extrados and corresponding compressive stresses at the intrados, consistent with theoretical expectations. These values were compared against ASME B31.1 allowable stress limits to confirm structural integrity.

Design Optimization of High-Pressure Turbine and Compressor Blade

Pratt & Whitney Canada

- **Objective:** Optimize aerodynamic and structural performance of high-pressure turbine and centrifugal compressor blades under stress, fatigue, and durability constraints.
- **Solution:** Used Python-based numerical optimization to explore design space and refine blade geometry. Modeled 3D blade profiles in CATIA V5 and conducted CFD and structural analysis in ANSYS Fluent to assess life cycle and off-design behavior.
- **Result:** Delivered an optimized blade design that met stress, material durability, and aerodynamic efficiency targets, enhancing performance and extending component life.

Design of Low emission Fuel Injection System for Gas Turbine

Concordia University

- **Objective:** Designed a low-emission hydrogen-fueled injector for gas turbines, minimizing NOx emissions while addressing operational needs and optimizing combustor performance.
- **Solution:** Incorporated micromixing and multipoint injection methods using ANSYS Fluent for CFD simulations and performance analysis. Performed extensive testing to validate the design criteria.
- **Result:** Resulted in a validated low-NOx injector supporting Siemens' goals in clean propulsion technology.

CFD Simulation and Performance Analysis of VKI High Pressure Turbine Blade

Concordia University

- **Objective:** Conduct 3D aerodynamic simulations of a VKI turbine blade cascade to analyze pressure distribution, flow-behavior, and performance sensitivity using ANSYS CFX and StarCCM+.
- **Solution:** Developed a parametric cascade model with structured mesh and applied realistic boundary conditions. Simulated compressible flow across various Mach numbers and analyzed key parameters like pressure coefficient, velocity, and isentropic Mach using post-processing tools.
- **Result:** Validated CFD models and extracted insights to support blade shape optimization and performance evaluation under off-design conditions.

Implementation of RQL (Rich Burn Quick Lean) Combustor in Cantera Modeling

McGill University

- **Objective:** To analyze the feasibility of hydrogen as an alternative fuel by blending it with jet fuel and visualizing the combustion dynamics of the blended fuel with varying proportions.
- **Solution:** Modeled hydrogen-blended jet fuel combustion using Cantera and Python, quantifying risks and demonstrating its feasibility to minimize harmful emissions and enhancing operational efficiency.
- **Result:** The simulation demonstrated that the mixture of hydrogen and jet fuel with optimum mixing ratio has the potential to significantly reduce NOx emissions, making hydrogen a viable alternative fuel option for cleaner combustion processes.

Certifications & Achievements

- Certification Training of Microsoft POWER BI for Project Management (2023)
- Selected for the prestigious Princeton University-CEFRC (Combustion Energy Frontier Research Center) Summer School on Combustion (2023)
- Aerodynamic CFD Simulation Training Course by ANSYS Fluent (In Progress)
- Finite Element Analysis using ANSYS APDL (In Progress)
- Short Term Training Program (STTP) On Pedagogy and Research Methodology Sponsored by GUJCOST (2021)
- CATIA V5 Complete Training – Udemy
- Active member of ASME under student membership program
- How to write & publish Scientific paper (Project-Centered Course)(2020)
- MATLAB Certification from UDEMY (2022)

Research Paper

Conducted research on micromix combustion as a low-NOx strategy for gas turbines by analyzing the aerodynamic behavior of hydrogen flamelets. Focused on flame structure, stabilization, and jet-in-crossflow interactions to assess flame anchoring, temperature uniformity, and NOx reduction. Compared micromix performance with DLN (Dry Low NOx) and RQL (Rich Burn Quick Lean) technologies from an aero-thermal standpoint, highlighting benefits in stability, mixing efficiency, and pressure loss control.